



Mammal goes extinct due to us

The Bramble Cay melomys, a rodent species, is the first mammal to go extinct because of human-caused climate change. <https://digit.in/mar19-10>



Methane fed crabs

Researchers have discovered a group of tanner crabs vigorously feeding at a methane seep on the seafloor. <https://digit.in/mar19-11>

Alkaloids

The mystery molecules



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hen you have a cup of tea or coffee, you are consuming caffeine and theobromine, which are both

alkaloids. A bar of chocolate gets its characteristic bitter taste because of the presence of theobromine. Generally speaking, alkaloids are a group of organic compounds with a high nitrogen content, and that originate mostly in plants. Many alkaloids have pronounced physiological effects on human beings.

The German physicist Walther Meissner gave the group its name in 1819, because they appeared to be “like alkali”. The group includes a large number of molecules, and their function within the organisms that produce them is not yet completely understood. The very definition of the word does not yet have a scientific consensus.

Due to the mysteries surrounding their function in both the organisms of their origin and the effect on humans, alkaloids are the subject of intense and sustained scientific research. Over the last ten years, there have been over 10,000 research papers on alkaloids. The research involves the discovery of new alkaloids (over 16,000 are already known), their role within the organisms that produce them, and the effects on the organisms that ingest them. Researchers are also producing entirely synthetic alkaloids in laboratories, which do not exist in nature, or at least have not been discovered yet.

The exact functions of the alkaloids in the plants that produce them remains an unsolved problem for science. The research takes long periods of time, the new molecules discovered often have unexpected structures, and are bioactive in surprising ways. While we may not have understood what they do in the plants, their effects on humans are better understood.

Humans have been using alkaloids for thousands of years, and they are the most physiologically active molecules of plant origin known. They have potential future uses in medicine, biotechnology and agriculture. Researchers believe that these molecules may be the key to finding cures for deadly diseases such as AIDS and cancer.

OCCURRENCE IN NATURE

About 25 percent of all plants have alkaloids of some form or another. Oranges, peas, peanuts and lentils are some of the edible plant groups that have alkaloids within them. Lilies, asters, nightshades and buttercups are some other plant groups with alkaloids in them.

Once extracted from the plants, most alkaloids are white solids. Some alkaloids with a low molecular weight, such as the arecoline found in the areca nut, are in a liquid form. Alkaloids with no oxygen atom in their structures, such as nicotine, are also in